

Silicon waveguide group index using Mach–Zehnder interferometers

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Public edX username: [confirm; design filename uses yihua]

Design report draft • 5 October 2026. Optics Express manuscript style. Experimental results and unfinished simulations are explicitly marked. This document does not claim experimental validation.

Abstract. A silicon photonic circuit is designed to determine waveguide group index from the free spectral range of asymmetric Mach–Zehnder interferometers. The supplied GDS layout contains five interferometers using Y branches, a broadband directional coupler, or multimode interference couplers, together with a reference waveguide. The layout specifies 500 nm wide TE waveguides and nominal 5 μm bend radii. Arm-length differences extracted from stored waveguide metadata range from 100.000 to 343.953 μm . Existing modal plots provide preliminary dispersion estimates. Circuit simulations, fabrication details, and measured spectra remain to be added. The report defines the extraction procedure and the checks needed before comparison with experiment.

1. Introduction

Group index determines optical group delay and the spectral spacing of interference fringes. An asymmetric Mach–Zehnder interferometer (MZI) converts the phase difference accumulated along two waveguide paths into a wavelength-dependent transmission. Measuring its free spectral range (FSR) enables extraction of group index when the optical path-length difference is known [1]. This project develops and compares several interferometer layouts for that purpose.

The intended platform is the course silicon-on-insulator process with a nominal 220 nm silicon device layer. The GDS explicitly identifies strip TE waveguides at 1550 nm with a width of 500 nm; the layer thickness is a course assumption and is not encoded by the two-dimensional mask. The final report will compare a simulated dispersion model with group indices extracted from measurements.

2. Interferometer theory

For a waveguide with effective index n_{eff} , the propagation constant and group index are

$$\beta(\lambda) = 2\pi n_{\text{eff}}(\lambda)/\lambda, \quad n_g = n_{\text{eff}} - \lambda \, dn_{\text{eff}}/d\lambda. \quad (1)$$

For ideal balanced splitters, one output transmission can be written using arm field attenuations A_1, A_2 and relative phase $\Delta\phi$ as

$$T = [A_1^2 + A_2^2 + 2A_1A_2 \cos(\Delta\phi)]/4, \quad (2)$$

$$\Delta\phi = \beta\Delta L + \phi_c. \quad (3)$$

The coupler phase ϕ_c and its dispersion depend on the selected ports. Equation (2) is an ideal reference model; the implemented Y–BDC and MMI circuits require component scattering

matrices rather than an assumption of identical ideal couplers.

2.1. Group index extraction

Successive maxima of the same output differ in phase by 2π . For identical waveguide cross sections and negligible differential component delay, their local frequency and wavelength spacings satisfy

$$\text{FSR}_f \approx c/(n_g|\Delta L|), \quad \text{FSR}_\lambda \approx \lambda^2/(n_g|\Delta L|). \quad (4)$$

$$n_g \approx c/(|\Delta L| \Delta f), \quad \Delta f = |c/\lambda_{m+1} - c/\lambda_m|. \quad (5)$$

Use adjacent maxima or adjacent minima, not a maximum–minimum pair. Assign each estimate to the midpoint frequency of that fringe pair. Fit multiple fringes to reduce peak-location error and compare results across devices. Differential coupler delay and unequal bend contributions must be assessed before interpreting the extracted value as the straight-waveguide group index.

3. Mask layout and device inventory

The supplied file *EBeam_yihua.gds* contains 34 cells with a 1 nm database unit. The physical top cell is *EBeam_design_yi*. The separate context-information cell is excluded from the optical mask rendering in Fig. 1. Three MZIs use a Y-branch input and a broadband directional coupler (BDC) output; two additional MZIs use paired Y branches or paired 2×2 MMI couplers. TE grating couplers provide fiber access. A separate reference route has a stored length of 223.710 μm .

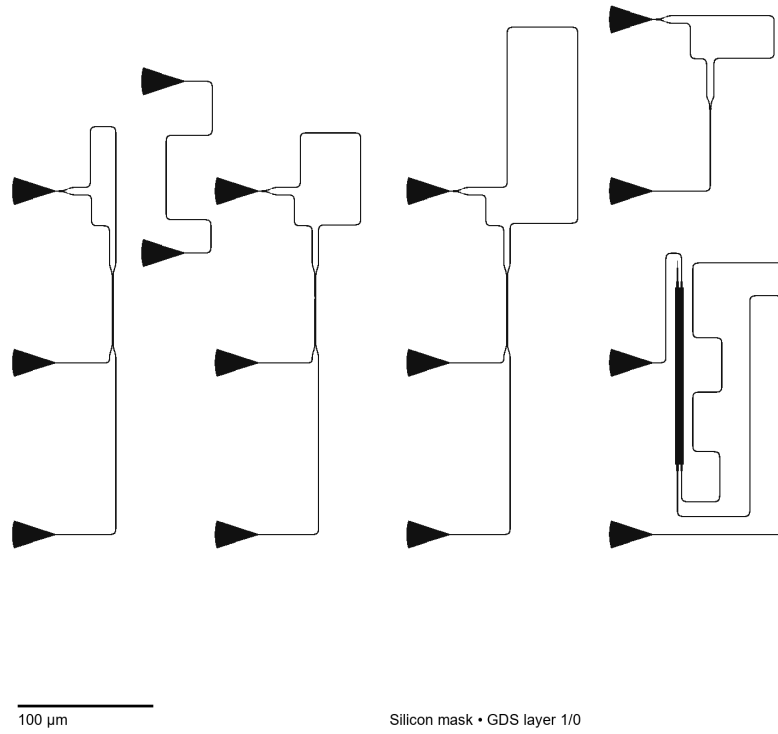


Fig. 1. Actual silicon-mask polygons flattened from the physical top cell of the supplied GDS, layer 1/datatype 0. The scale bar is 100 μm . The figure omits annotation and non-silicon layers and is a layout visualization, not a verification result.

3.1. Arm lengths and expected spectral spacing

Table 1 uses the waveguide lengths stored in the GDS component metadata. Lengths are between the associated component ports; external fiber-access routes are excluded from the arm difference. These values still require confirmation against the final extracted netlist and geometric path lengths.

Table 1. Stored arm lengths and preliminary analytical FSR at 1550 nm

Cell	Splitter / combiner	Short μm	Long μm	ΔL μm	FSR nm
MZI100	Y branch / BDC	69.745	169.745	100.000	5.716
MZI150	Y branch / BDC	69.745	219.746	150.001	3.811
MZI350	Y branch / BDC	69.745	413.698	343.953	1.662
MZI_Y	Y branch / Y branch	71.055	171.055	100.000	5.716
MZI_MMI	MMI / MMI	269.153	369.153	100.000	5.716

The FSR estimates assume $n_g = 4.203$, approximately read from the existing plot, and ideal component phase behavior. They are analytical design estimates, not completed circuit simulations. All are below the approximately 50 nm measurement bandwidth specified in the course; the smallest expected spacing is approximately 1.662 nm, well above a 1 μm minimum laser step.

Length discrepancy requiring review. The cell called *MZI350* stores arm lengths of 69.745 and 413.698 μm , giving $\Delta L = 343.953 \mu\text{m}$. Substituting 350 μm into Eq. (5) would bias the inferred group index low by approximately 1.73% if the stored lengths are correct. Confirm the centerline lengths in KLayout/SiEPIC before final simulation and extraction; the source GDS has not been changed.

The waveguide metadata specifies a nominal 5 μm bend radius. The layout contains a saved SiEPIC verification annotation reporting zero errors, SiEPIC-Tools v0.5.31, EBeam v0.4.53, and KLayout v0.30.12. This is an annotation embedded in the file; design-rule and connectivity checks have not been rerun for this report.

4. Waveguide modelling

A two-dimensional eigenmode calculation supplies the TE mode profile and wavelength-dependent effective index. Existing local PNG exports are reproduced in Figs. 2 and 3. They are preliminary simulation evidence; the solver configuration, material model, cladding, mesh convergence, and numerical arrays still need to be documented. Approximate values at 1550 nm are $n_{\text{eff}} \approx 2.45$ and $n_g \approx 4.203$, as read from the plotted curves.

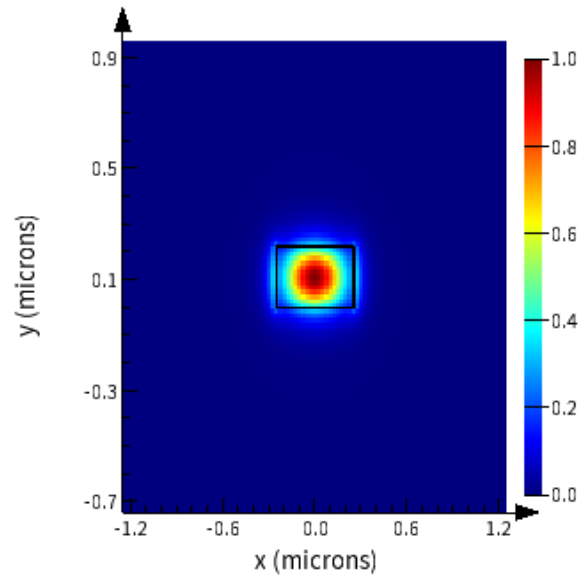


Fig. 2. Existing local mode-profile export, Mode_profile_TE_polarization_model.png. Add the field quantity, units, cross-section dimensions, and solver settings after checking the source simulation.

4.1. Dispersion and compact model

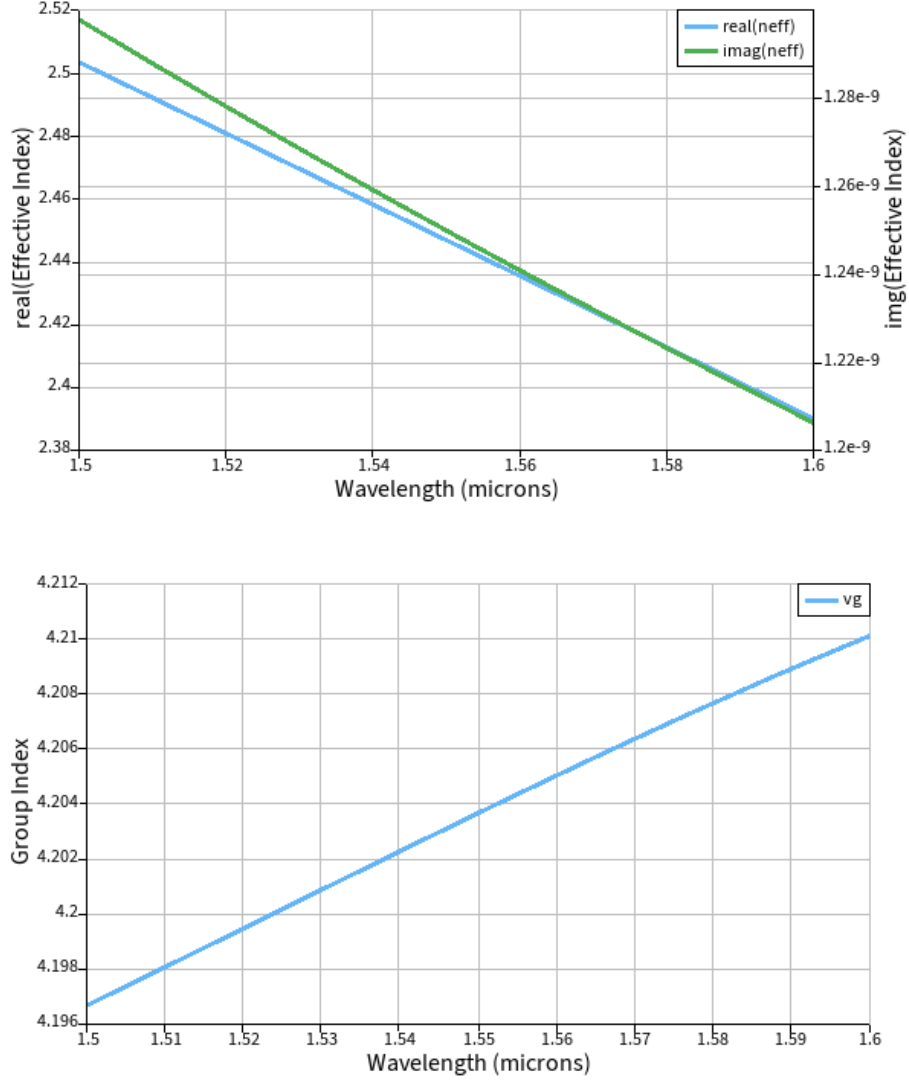


Fig. 3. Existing local effective-index (top) and group-index (bottom) exports for the nominal 500 nm waveguide. Values are not re-derived here from raw solver data. The group-index curve and its axis labels must be checked against the numerical export before the final report.

Fit the real effective index around $\lambda_0 = 1.55 \mu\text{m}$ to a quadratic polynomial, with wavelength consistently expressed in micrometres:

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2. \quad (6)$$

To be completed. Insert numerical coefficients a_0 , a_1 (μm^{-1}), a_2 (μm^{-2}), wavelength range, fit residuals, and mesh-convergence results. Derive group index from Eq. (1) and compare with the solver export. Document loss separately; a small imaginary effective index alone is not a measured propagation loss.

5. Circuit simulations to complete

Use the verified layout connectivity and port-to-port lengths to construct each interferometer in a circuit simulator. A dispersive waveguide model and wavelength-dependent component scattering parameters are needed to predict both fringe spacing and visibility [2].

5.1. Component and bend models

To be completed. Confirm compatible EBeam compact models for the TE grating couplers, Y branches, BDC, and terminators. For the custom MMI_2x2 cell, locate a validated model or perform an EME/FDTD wavelength sweep. Insert its geometry, mesh convergence, complex S-parameters, insertion loss, splitting ratio, and relative phase. Check whether 5 μm bends introduce differential loss or group delay significant for the extraction.

5.2. Interferometer transmission

To be completed. Insert simulated transmission versus wavelength for all five devices, covering the intended laser scan range. Plot both accessible outputs of each Y–BDC MZI. State source polarization, port mapping, spectral sampling, component-model versions, and waveguide loss assumptions. Add a simulated-spectrum figure here once those calculations are complete.

5.3. Validate the extraction method

To be completed. Apply the same fringe-fitting routine intended for measurements to the simulated spectra. Tabulate extracted FSR and group index for each device, then compare with the eigenmode group index. Quantify any bias caused by coupler dispersion, bends, or spectral sampling. Repeat for reasonable width/thickness variations using documented process tolerances.

6. Fabrication and experimental method

To be completed. Record fabrication provider, silicon and oxide thicknesses, cladding, etch process, fabrication date, and any available dimensional inspection. Replace the nominal process assumptions with confirmed values.

The planned experiment couples TE-polarized light through the on-chip grating couplers and sweeps a tunable laser over the available wavelength range. Record a reference spectrum through the dedicated route and the output spectrum of each interferometer. Document power, wavelength calibration, scan increment, polarization adjustment, detector settings, alignment, temperature, and repeatability. Reference normalization may reduce the grating-coupler envelope, but its route loss and alignment differences must be considered.

To be completed. Insert an experimental setup diagram and the actual instrument settings. No measurement results are currently available.

7. Measurement results and analysis

To be completed. Insert raw and normalized measured spectra, including repeat scans and the reference transmission. Report fringe visibility and usable wavelength range. Identify rejected fringes and the reason for rejection.

Table 2. Measured values to be filled after testing

Device	λ range	Measured FSR	Extracted n_g	Uncertainty
MZI100	—	—	—	—
MZI150	—	—	—	—
MZI350	—	—	—	—
MZI_Y	—	—	—	—
MZI_MMI	—	—	—	—

For independent uncertainties in fringe spacing and path-length difference, a first approximation is

$$(\sigma_{n_g}/n_g)^2 \approx (\sigma_{\Delta f}/\Delta f)^2 + (\sigma_{\Delta L}/|\Delta L|)^2. \quad (7)$$

Include wavelength calibration and peak-fitting error in the spacing uncertainty. Report correlated calibration effects and model biases separately rather than treating them as independent random errors. Compare measurements with both eigenmode predictions and circuit-derived values at matching wavelengths.

8. Discussion and conclusion

The mask provides multiple arm differences and three splitter/combiner architectures for studying group-index extraction. The present inventory establishes the design geometry and identifies a length-label discrepancy that must be resolved. Existing modal plots support a preliminary design estimate, but completed circuit simulations and experimental spectra are required for a quantitative conclusion.

To be completed. Discuss simulation–measurement agreement, uncertainty, bend and coupler effects, and differences between device architectures. State the final measured group index with its wavelength range and uncertainty. Do not claim agreement or successful fabrication until supported by results.

Funding

To be completed. Add funding information if applicable.

Acknowledgments

To be completed. Identify course, fabrication, software, and other assistance as appropriate.

Disclosures

To be completed. Add the author’s confirmed disclosure statement.

Data availability

The design source is EBeam_yihua.gds supplied by the author. Figures 2–3 come from local report_simulation PNG exports. Numerical simulation tables and measurement datasets remain to be added. [Insert the final repository or data-availability statement.]

References

1. L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems* (Cambridge University Press, 2015), doi:10.1017/CBO9781316084168.
2. Ansys, “INTERCONNECT product reference manual,” Ansys Optics documentation (accessed 5 October 2026).

Draft provenance and completion notes

This draft follows the single-column manuscript conventions described in the Optica style guide and author template guidance. It is not a published article. Remove this completion-note section from the final submission.

Before submission, confirm the author information and required EBeam_USERNAME.pdf filename; verify the final GDS/netlist lengths and current design checks; complete the dispersion coefficients, component models, and circuit spectra; and replace all pending measurement and analysis sections. Preserve the distinction between numerical simulations, analytical estimates, stored layout metadata, and experimental observations.